Status	Finished
Started	Friday, 9 October 2026, 11:20 AM
Completed	Friday, 9 October 2026, 11:24 AM
Duration	3 mins 20 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Linear search ✓
- b. Interpolation search
- c. Hash search
- d. Binary search

Question **2**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Searching ✓
- b. Discovering
- c. Finding
- d. Mining

Question **3**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n/2
- c. log n
- d. n-1

Question **4**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. n ✓
c. log n
d. 1

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It requires the list to be sorted ✓
c. It does not work with negative numbers
d. It can only be applied to large lists

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is unsorted
c. When the list is very small
d. When the list is sorted ✓

Question **7**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The search continues in the right sublist
c. The list is sorted
d. The search continues in the left sublist

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search continues from the beginning
- c. The list is sorted
- d. The search stops

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search is more efficient than binary search on large lists.
- b. Linear search requires the list to be sorted.
- c. Linear search divides the list into halves.
- d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. Number values should only be present
  Number values should only be present
- b. There should be direct access to the middle element in any sublist
- c. There must be a mechanism to delete and/or insert elements in the list ✓
- d. The list must be sorted

Question **11**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 94
- b. 90 and 99 ✓
- c. 89 and 99
- d. 89 and 94

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Inserting elements into a list
- c. Sorting elements in a list
- d. Deleting elements from a list

Question **13**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the first element in the list
- b. Compare the target element with the middle element in the list ✓
- c. Divide the list into two equal parts
- d. Sort the list

Question **14**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Hash
- b. Both (1) & (3)
- c. Linear
- d. Binary ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Hash search
- b. Linear search
- c. Binary search ✓
- d. Sequential search